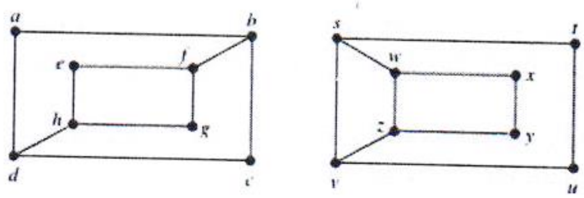
19/11/19

Indian Institute of Engineering Science and Technology, Shibpur
B.Tech. (CST) 3rd Semester Final Examinations, November 2019
Discrete Structures (CS - 303)

Time: 3 hours **Full Marks: 70**

Answer *Q.1* and *any five* from the rest
(Write all parts of the same question together)

1. Give short answers to the following: [2x10 = 20]
   a) A compound proposition that is always __________ is called a contradiction, and the one which is neither a tautology nor a contradiction is called a __________.
   b) What are the *contrapositive* and *inverse* of the conditional statement "I come to class whenever there is going to be a test."?
   c) Two sets A and B contains a and b elements respectively. If power set of A contains 16 more elements than that of B, what are the values of a and b?
   d) A function is defined by mapping $f: A \rightarrow B$ such that A contains m elements and B contains n elements and $m \leq n$. Find the number of one-to-one functions.
   e) Are the following statements *True* or *False*?
      i. An equivalence relation has reflexive, anti-symmetric, and transitive properties.
      ii. A partially ordered relation has reflexive, symmetric, and transitive properties.
   f) Write any two properties of a simple graph.
   g) A pseudo-graph has __________ and/or __________ connecting the same vertex pair.
   h) How many edges are there in a graph with 10 vertices each of degree six?
   i) What is the trade-off between *adjacency list* and *adjacency matrix* representation of graphs?
   j) Determine whether the following graphs (G & H) are isomorphic.



G *H*

2. a) Use rules of inference to show that the hypotheses "If it does not rain or if it is not foggy, then the sailing race will be held and the lifesaving demonstration will go on," "If the sailing race is held, then the trophy will be awarded," and "The trophy was not awarded" imply the conclusion "It rained."

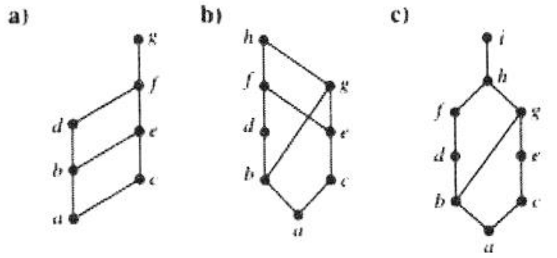
   b) Suppose the relation on a set R is represented as $M_R = \begin{bmatrix} 1 & 1 & 0 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$. Determine if this relation is *reflexive, irreflexive, symmetric, anti-symmetric, and transitive* (give reasons). Use Boolean multiplication of zero-one matrix wherever necessary.

[5 + (1x5) = 10]

3. a) Prove that the transitive closure of a relation R equals the connectivity relation R^*.
   b) Use Warshall's algorithm to find the transitive closure of the relation $R = \{(2,1), (2,3), (3,1), (3,4), (4,1), (4,3)\}$ on the set $S = \{1, 2, 3, 4\}$.

[4 + 6 = 10]

4. a) Draw the *Hasse diagram* for the "less than or equal to" relation on {0, 2, 5, 10, 11, 15}. Find the maximal, minimal, greatest, and least elements in the diagram.
   b) Determine whether the posets with the following Hasse diagrams are lattices (give reasons).



[(2 + ½ x4) + (2x3) = 10

5. a) State the *Fundamental Theorem of Arithmetic.*
   b) "There are infinitely many primes" – Use Euclid's method to prove this theorem.
   c) What is *Bézout's identity*? Express $gcd\ (252,\ 198) = 18$ as a linear combination of 252 and 198.

[2 + 4 + (1 + 3) = 10]

6. a) Triangulation is the process of dividing a simple polygon into triangles by adding nonintersecting diagonals. Use strong induction to prove that a simple polygon with n sides, where n is an integer with $n \geq 3$, can be triangulated into $n - 2$ triangles. *(hint: "Every simple polygon with at least four sides have an interior diagonal")*
   b) Use mathematical induction to prove that $n^2 - 1$ is divisible by 8 whenever n is an odd positive integer.

[5+5 = 10]

7. a) Suppose that a weapons inspector must inspect each of five different sites twice, visiting one site per day. The inspector is free to select the order in which to visit these sites, but cannot visit site X, the most suspicious site, on two consecutive days. In how many different orders can the inspector visit these sites?
   b) Use *Generalize Pigeonhole Principle* to show that among any set of 5 integers, there are 2 with the same remainder when divided by 4.
   c) How many solutions does the equation $x_1 + x_2 + x_3 = 11$ have, where the variables are the integers with $x_1 >= 1$, $x_2 >= 2$ and $x_3 >= 3$?

[4 + 3+ 3 = 10]